Seungmin Cha

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EDUCATION

- **Yonsei University**, Seoul, South Korea — Mar 2022 – Present

  Major in Physics; Double Major in Intelligent Semiconductor Technology

  Minor in Applied Statistics (in progress)
  - Relevant coursework: Quantum Mechanics, Electromagnetics, Materials Science, Intelligent Semiconductor Systems, Machine Learning, Deep Learning, Quantum Machine Learning, Statistical Mechanics, Thermodynamics, and Linear Algebra

RESEARCH INTERESTS

- **Advanced Semiconductor Packaging:** Heterogeneous Integration, High-Bandwidth Memory (HBM), 2.5D/3D Integration, Interconnect Technologies, Thermal Management
- **Condensed Matter Physics:** Semiconductor Physics, 2D Materials, Magnetism, Optical Spectroscopy
- **Quantum Computing:** Quantum Machine Learning, Superconducting Qubits, Circuit QED, Quantum Error Correction

RESEARCH EXPERIENCE

- **Undergraduate Researcher**, Quantum Optics Lab, Yonsei University — Mar 2026 – Jun 2026
  - Advisor: Prof. Seongjin Hong
  - Studied and implemented the active optical intensity interferometry framework presented in *Active Optical Intensity Interferometry* (Phys. Rev. Lett., 2025).
  - Developed simulation code and performed data processing and analysis for correlation-based image reconstruction in optical interferometry.
  - Participated in optical experiments to test and reproduce interferometric imaging schemes for long-baseline, high-resolution sensing.
  - Documented research progress, simulation workflows, and literature reviews in a personal archive.
- **Undergraduate Researcher**, Quantum Information Theory Lab, Yonsei University — Jan 2024 – Jun 2024
  - Advisor: Prof. Kyungsun Moon
  - Studied the theoretical foundations of quantum machine learning through the textbook *Machine Learning with Quantum Computers* and prepared presentation slides for lab seminars.
  - Read and reviewed *Supervised Learning with Quantum-Enhanced Feature Spaces* (Nature, 2019), with a focus on quantum feature maps, kernel estimation, and classification with support vector machines.
  - Presented and discussed core concepts in quantum-enhanced supervised learning during lab meetings.
  - Maintained written study notes and summaries on quantum computing topics on a personal archive.

PROJECTS

- **Tilted-Array HBM (taHBM)**, STOB League — Jul 2026 – Present
  - Co-developed a tilted-die memory packaging concept that uses a patterned support substrate and redistribution layers (RDLs) to connect individual DRAM dies without through-silicon vias (TSVs) or a base die.
  - Investigated a fabrication approach combining anisotropic silicon etching, inclined-surface metallization, and indium microbump bonding for die placement and electrical interconnection.

- Performed preliminary geometric and interconnect calculations to assess die pitch, routing-length differences, and signal propagation skew.
- Explored inter-die cooling channels for thermal management and examined fabrication and assembly constraints.
- Preparing invention disclosure documents and technical materials for a patent application on the proposed memory packaging architecture.

WORK EXPERIENCE

- **SK hynix Ambassador** Apr 2026 – Oct 2026
  - Received the **Outstanding Ambassador Award**.
  - Created multiple semiconductor-focused social media contents communicating SK hynix's technologies, corporate vision, and industry insights from a student perspective.
  - Hosted and conducted interviews with semiconductor experts in SK hynix's official YouTube educational series, "Future Talent CLASS," featuring discussions on next-generation memory technologies, including HBM with Prof. Changhwan Shin (Korea University), and PIM with Prof. Sebum Paek (KAIST), while translating complex semiconductor concepts into accessible educational content for students and the general public.
  - Participated in SK hynix's "N-Job Explorer" series, interviewing System Architecture & Software Solution (SA&SS) engineers, experiencing core job responsibilities firsthand, and communicating the role and workplace environment to a broader audience.
  - Participated in a monthly ambassador program featuring semiconductor and AI trend lectures, networking with industry professionals, and other professional development sessions.
- **Signals Intelligence (SIGINT) Specialist**, Republic of Korea Army Aug 2024 – Feb 2026
  - Served as a Signals Intelligence specialist and Squad Leader in the Republic of Korea Army.

AWARDS AND SCHOLARSHIPS

- **Military Honor Family Designation** (Byeongyeok Myeongmunga) 2026
  - Recognized as a distinguished military family for three consecutive generations of honorable active-duty service.
- **Scholarship for Outstanding STEM Talent**, KC Scholarship Foundation 2024
- **5th Place**, DB GAPS Investment Competition 2023
- **Minister of Science and ICT Award** 2013

CERTIFICATIONS

- **ADsP** (Advanced Data Analytics Semi-Professional) 2025
- **Computer Literacy Level 2** 2025
- **SQLD** (SQL Developer) 2023

ACTIVITIES

- **Bookmark** (Yonsei University Library Student Council) Sep 2026 – Present
  - Participating in library support and student volunteer activities.
- **RU:SH** (Runners United: Sinchon) Feb 2026 – Present
  - Running crew with students and alumni from Yonsei, Ewha, and Sogang Universities
- **QIYA** (Quantum Informatics at Yonsei Academy) Jan 2026 – Present

  - Completed the *Basic Quantum Information Theory* course and IBM Learning courses as part of structured study in quantum information science.
  - Wrote regular reports and delivered presentations on course materials and related topics in quantum information.

- **Completion of the 5th Quantum Information Science Winter School** Jan 2026
  - Program organized by the Korean Physical Society (KPS) and the Quantum Information Society of Korea (QISK).
- **Kwang** (CrossFit Club at Yonsei University) Mar 2023 – Present
- **Gongsarang** (Basketball Club at Yonsei University) Mar 2022 – Dec 2022

ADDITIONAL INFORMATION

- **Languages:** Korean (Native), English (Working Proficiency)
- **Technical Skills:** Python, Qiskit, SQL, Data Processing, Data Analysis
- **Interests:** Running (10K: 40:49, Half Marathon: 1:34:04, Marathon 3:29:42), CrossFit, Reading, and Film